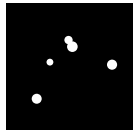
Module 2 - stage 2 fine-tuning step (train_module2_poc.py)

initialised from stage 1
(final-G.ckpt / final-D.ckpt)



X
baseline image
(3ch) + Module 1
mask (1ch)



concat, 4+c_dim ch

Generator



G(x)
synthesized
image, target
stage (3ch)

after every epoch:
L1(G(x), Reg(Y)) on 10 %
held-out eyes / patients
lowest → best-G.ckpt
(50 epochs)

Discriminator

d_loss = fake − real + 10·GP
g_loss = adv + λ_rec·L1(G(x), Reg(Y))
λ_rec = 10, λ_adv = 1

c target stage, one-hot
(= follow-up grade)

s = c (AdaIN style vector)



Y
real follow-
up (raw)

Affine registration



Reg(Y)
follow-up
warped to
X's geometry

L1
rec

pairs: FIRE + LongDR + Tianjin; registration net trained once, self-supervised (photometric loss)